Overlap of stool-decreased cellulose-annotated taxa against six null models

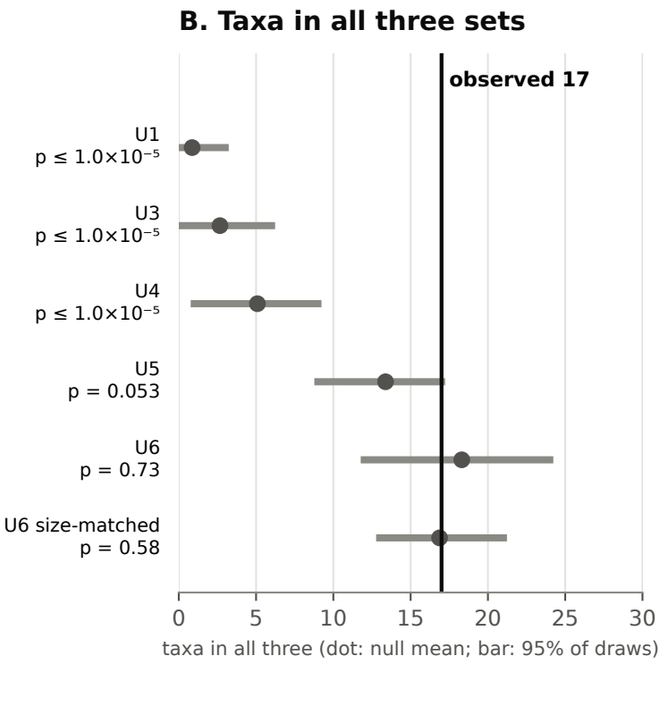
Nulls that draw taxa independently (U1–U4) make the overlap look extreme; nulls that respect how often taxa are reported (U5) and which taxa are reported together (U6) do not.

A. Expected Jaccard distance and p under each null

	observed d_J	U1 matrix uniform	U2 stool uniform	U3 stool dec uniform	U4 random trait	U5 prevalence weighted	U6 signature permutation	U6 size-matched
PD vs. HIV	**0.488**	0.914 exact 1.1×10^{-14}	0.876 exact 1.8×10^{-11}	0.841 exact 2.5×10^{-9}	0.795 p 3.0×10^{-5}	0.621 p 0.024	0.550 p 0.21	0.541 p 0.20
PD vs. CRC	**0.400**	0.894 exact 1.0×10^{-17}	0.845 exact 1.6×10^{-13}	0.799 exact 1.1×10^{-10}	0.747 p $\le 1.0\times10^{-5}$	0.546 p 0.019	0.491 p 0.10	0.462 p 0.18
HIV vs. CRC	**0.550**	0.918 exact 2.6×10^{-11}	0.882 exact 1.2×10^{-8}	0.848 exact 6.8×10^{-7}	0.783 p 0.001	0.622 p 0.19	0.540 p 0.55	0.537 p 0.58
Mean of 3 pairs	**0.479**	0.909 p $\le 1.0\times10^{-5}$	0.868 p $\le 1.0\times10^{-5}$	0.829 p $\le 1.0\times10^{-5}$	0.775 p $\le 1.0\times10^{-5}$	0.596 p 0.005	0.527 p 0.19	0.513 p 0.24

Observed values are in the first column. Shaded blue: $p < 0.05$. U1–U3 pair p-values are exact hypergeometric tails; all other p are Monte Carlo. U4 places a random trait on the taxa reported decreased in stool in the three conditions; U5 weights taxa by their reporting frequency in other stool studies; U6 permutes whole study signatures.

B. Taxa in all three sets



C. The 47 taxa by the conditions that report them decreased in stool

PD, HIV and CRC 17	PD and CRC only 10	PD and HIV only 4	HIV and CRC only 1	PD only 7	CRC only 6	HIV only 2
Lachnospiraceae	Roseburia faecis	Bacteroidales	Actinomycetota	Eubacteriaceae	Blautia obeum	Bacillus
Roseburia	Anaerostipes hadrus	Bacteroidaceae		Gammaproteobacteria	Hominimerdicola alba	Carnobacteriaceae
Blautia	Bacilli	Lachnoclostridium		Lacrimispora	Lachnospira multipara	
Clostridium	Comamonadaceae	Mediterraneibacter torques		Mediterraneibacter	Pseudomonadaceae	
Bacillota	Lactobacillales			Micrococcaceae	Ruminococcus champanellensis	
Bacteroides	Bacteroides cellulosilyticus			Sphingobacteriaceae	Serratia	
Clostridia	Pseudomonadota			Streptomycetaceae		
Lachnospira	Ruminococcus bromii					
Oscillospiraceae	Ruminococcus flavefaciens					
Ruminococcus	[Eubacterium] siraeum					
Clostridiaceae						
Eubacteriales						
Anaerostipes						
Bacteroidia						
Bacteroidota						
Eubacterium						
Butyrivibrio						

Names are KG-Microbe matrix taxon names; family-and-above nodes carry annotations rolled up from descendants. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.